MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OFAERONAUTICAL AND AUTOMOBILE ENGINEERING			
END SEMESTER EXAMINATION		**NOVEMBER, 2024**	
Level	**B.Tech**	**Curriculum**	**2018**
Program Name	**AUTOMOBILE ENGINEERING**	**Course Type**	**PE**
Course Name	**Farm Machinery and Earth Moving Equipment**	**Course Code**	**AAE 4078**
Semester	**VII**	**Max Marks**	**50**
Date	**20-11-2024**	**Duration**	**3 hours**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		IET		BL
			CO	PO	CLO	AHEP 4 LO	
1A	Sketch the layout of tower crane and explain the features of the important components and working of the same.	**03**	CO1	1,2,3,4, 5,6,7,9, 12	**1**	1,2,3,5, 12,18	**3**
1B	Classify the dredger system based on application and interpret the working of suction dredger unit.	**03**	CO2	1,2,3,4, 5,6,7,9, 12	**2**	1,2,3,5, 12,18	**3**
1C	Categorize the loader buckets based on applications and explain the features any three of them.	**04**	CO2	1,2,3,4, 5,6,7,9, 12	**2**	1,2,3,5, 12,18	**3**
2A	Identify the excavators based on the propulsion system and describe the features of any three of them.	**04**	CO1	1,2,3,4, 5,6,7,9, 12	**1**	1,2,3,5, 12,18	**3**
2B	With suitable layout, explain the use of Ward-Leonard System in operating/controlling the electric shovels.	**03**	CO1	1,2,3,4, 5,6,7,9, 12	**1**	1,2,3,5, 12,18	**4**
2C	Differentiate between the wheel loader and crawler loader based on its construction features.	**03**	CO2	1,2,3,4, 5,6,7,9, 12	**2**	1,2,3,5, 12,18	**2**

3A	Sketch a simplified layout of scraper unit and explain the working mode and operational features of the major components.	**05**	CO2	1,2,3,4, 5,6,7,9, 12	**2**	1,2,3,5, 12,18	**3**
3B	Discuss the effect of alttitude on 2 stroke and 4 stroke IC engine used in earth moving operations.	**03**	CO1	1,2,3,4, 5,6,7,9, 12	**1**	1,2,3,5, 12,18	**4**
3C	Anayalze the effect of rolling resistance and grade resistance on power requirement of a I C engine utilized in earth moving operations.	**02**	CO1	1,2,3,4, 5,6,7,9, 12	**1**	1,2,3,5, 12,18	**4**
4A	Classify the different types of sprayers used in agriculture field and with suitable layout explain the working and features of rocking Sprayer.	**04**	CO2	1,2,3,4, 5,6,7,9, 12	**2**	1,2,3,5, 12,18	**2**
4B	Explain the mode of power transmission system and cutting operation involved in a Mower or harvesting equipment.	**02**	CO1	1,2,3,4, 5,6,7,9, 12	**1**	1,2,3,5, 12,18	**3**
4C	Identify and elucitade the major safety measures need to be considered during a earth moving operation.	**04**	CO4	1,2,3,4, 5,6,7,9, 12	**4**	1,2,3,5, 12,18	**2**
5A	Identify the material handling systems based on applications and with neat sketch explain the working of Hydraulic Jack.	**04**	CO2	1,2,3,4, 5,6,7,9, 12	**2**	1,2,3,5, 12,18	**2**
5B	Ellobrate the objectives of preventive maintenance and explain the major preventive maintenance need to be performed in earth moving equipments.	**04**	CO3	1,2,3,4, 5,6,7,9, 12	**3**	1,2,3,5, 12,18	**3**
5C	Differentiate between the constant and variblae hydraulic delivery system.	**02**	CO1	1,2,3,4, 5,6,7,9, 12	**1**	1,2,3,5, 12,18	**2**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	**19**
2	**23**
3	**04**
4	**04**
5	

BL	Total Marks
2	**17**
3	**25**
4	**08**
5	